1 Problem Setup

> X_1, ..., X_n represents the fiber in the butternut group, and Y_1, ..., Y_n in the acorns.

We assume the subjects are normally distributed around our measure of interest (with the same variance). That is,

$$X \sim N(\mu_1, \sigma^2)$$
$$Y \sim N(\mu_2, \sigma^2)$$

The goal is to determine if acorns have a higher average fiber than butternuts.

2 C calculation

To accomplish this, we must first define "α":

> α represents the probability of the observed difference in sample means exceeding a specific value C, assuming that there truly is no difference in the mean fiber of acorns and butternuts. That is,
>
> $$\alpha = P\left(\overline{Y}_n - \overline{X}_n > C \,|\, \mu_2 = \mu_1\right)$$
>
> Note that C depends on the predetermined α-level. The remainder of this section focuses on calculating C.

Simplifying this probability requires us to utilize distributional properties of our sample means.
Note that the variance of a sample mean equals σ^2/n, and hence the variance of our difference is twice that:

$$\overline{Y}_n - \overline{X}_n \sim N(\mu_2 - \mu_1,\, 2\sigma^2/n)$$

Under the null hypothesis that $\mu_2 = \mu_1$, we have

$$\overline{Y}_n - \overline{X}_n \sim N(0,\, 2\sigma^2/n)$$

Probabilities under the standard normal distribution $N(0,\,1)$ have nice properties which make evaluation simpler. The next step in evaluating our probability is to convert it into terms of the standard normal.

Clarifying Example

Scaling a variable, which follows a normal distribution, by a quantity of σ multiplies its variance by σ^2.

For example, let $Z \sim N(0,\,1)$. Then,

$$\sigma Z \sim N(0,\,\sigma^2)$$

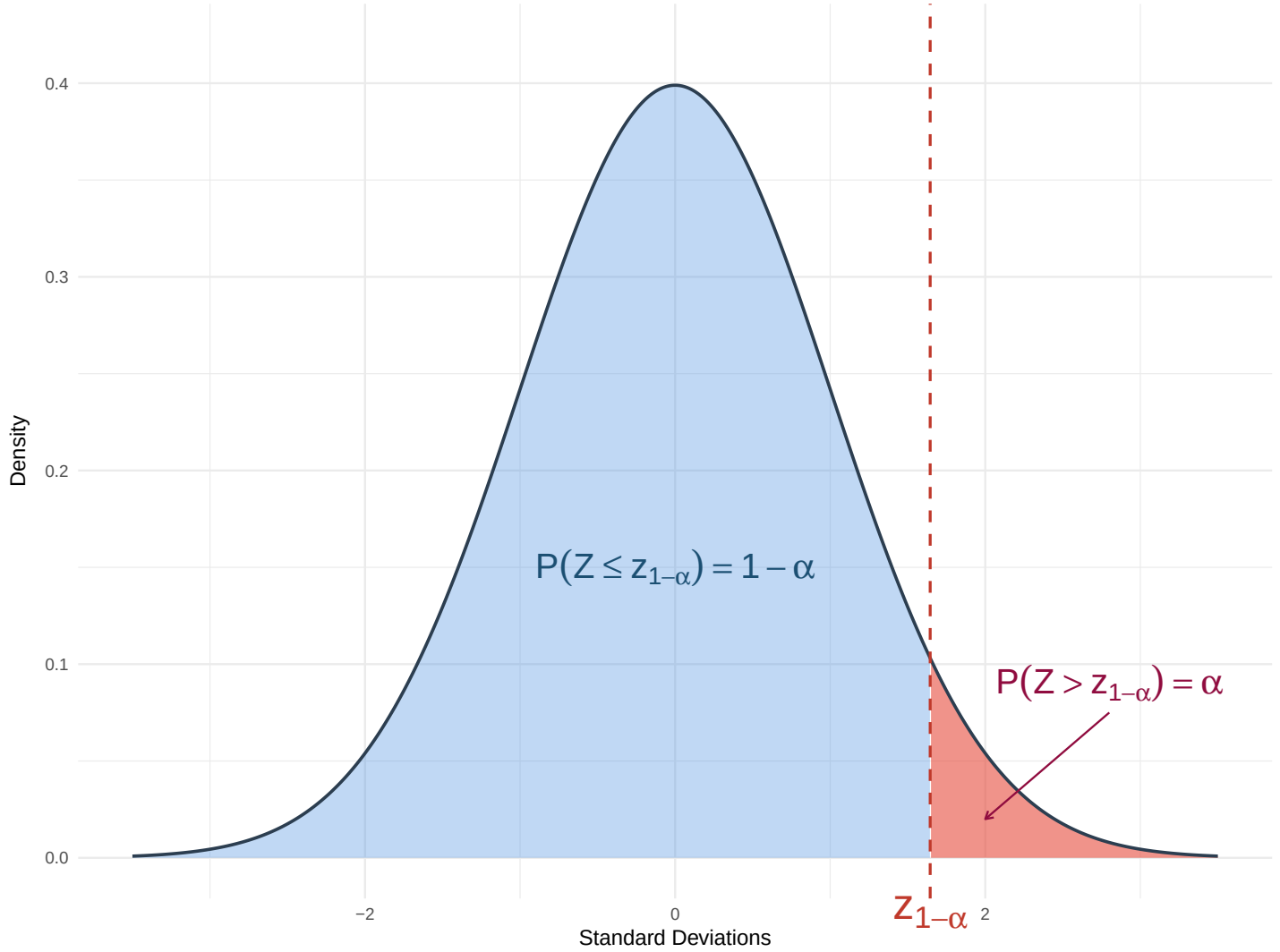
Utilizing this property:

$$\left(\overline{Y}_n - \overline{X}_n\right) \cdot \frac{\sqrt{n}}{\sigma\sqrt{2}} \sim N(0,\,1)$$

Now, our probability becomes

$$P\left(\overline{Y}_n - \overline{X}_n > C \,|\, \mu_2 = \mu_1\right) = P\left\{\left(\overline{Y}_n - \overline{X}_n\right) \cdot \frac{\sqrt{n}}{\sigma\sqrt{2}} > C \cdot \frac{\sqrt{n}}{\sigma\sqrt{2}}\right\}$$

We want the RHS of the inequality inside our probability to represent the $1-\alpha$ quantile for the probability to be equal to α. *(see figure)*


0.4
0.3
0.2
0.1
0.0
Density
$P(Z \leq z_{1-\alpha}) = 1 - \alpha$
$P(Z > z_{1-\alpha}) = \alpha$
−2
0
2
$z_{1-\alpha}$
Standard Deviations

Notationally, because $LHS = \left(\overline{Y}_n - \overline{X}_n\right) \cdot \frac{\sqrt{n}}{\sigma\sqrt{2}} \sim N(0,\, 1)$, we have

$$RHS = z_{1-\alpha} = C \cdot \frac{\sqrt{n}}{\sigma\sqrt{2}}$$

$$\implies C = z_{1-\alpha} \cdot \frac{\sigma\sqrt{2}}{\sqrt{n}}$$

That is,

$$\alpha = P\left(\overline{Y}_n - \overline{X}_n > z_{1-\alpha} \cdot \frac{\sigma\sqrt{2}}{\sqrt{n}} \,\middle|\, \mu_2 = \mu_1\right)$$

If we want there to be a 5% chance of observing the difference in mean fiber between acorns and butternuts found in the data, or more extreme differences, then we require $z_{1-\alpha} = z_{1-.05} = z_{0.95} \approx 1.645$.

For sake of discussion, if $\sigma = 1$ and $n = 8$, then there is a 5% chance that the difference exceeds ≈ 0.822.
If instead $n = 100$, then the value becomes ≈ 0.233.

3 Sample size calculation

Power $(1-\beta)$ is defined as the probability of rejecting the idea that butternuts and acorns have the same amount of fiber, when in reality acorns have more. I.e.,

$$1-\beta = P\left(\overline{Y}_n - \overline{X}_n > z_{1-\alpha} \cdot \frac{\sigma\sqrt{2}}{\sqrt{n}} \,\middle|\, \mu_2 > \mu_1\right)$$

The goal now is to find the sample size n in terms of α and β, which is accomplished by manipulating the probability above.

We standardize the LHS of the inequality, allowing us to write the RHS in terms of standard normal quantiles:

1. Subtract the difference in means by the true difference $(\mu_2 - \mu_1)$
2. Divide by the standard error $(\sigma\sqrt{2}/\sqrt{n})$

$$
\begin{aligned}
1-\beta &= P\left(\overline{Y}_n - \overline{X}_n - (\mu_2-\mu_1) > z_{1-\alpha}\cdot\frac{\sigma\sqrt{2}}{\sqrt{n}} - (\mu_2-\mu_1)\,\Big|\,\mu_2>\mu_1\right)\\
&= P\left(\frac{\sqrt{n}\left(\overline{Y}_n - \overline{X}_n - \mu_2 + \mu_1\right)}{\sigma\sqrt{2}} > z_{1-\alpha} - \frac{\sqrt{n}(\mu_2-\mu_1)}{\sigma\sqrt{2}}\Big|\,\mu_2>\mu_1\right)
\end{aligned}
$$

Because the LHS is standardized, the RHS is equal to z_β.

Now, we can find the sample size by solving for n:

$$z_\beta = z_{1-\alpha} - \frac{\sqrt{n}(\mu_2 - \mu_1)}{\sigma\sqrt{2}}$$

$$\frac{\sqrt{n}(\mu_2 - \mu_1)}{\sigma\sqrt{2}} = z_{1-\alpha} - z_\beta$$

$$\text{symmetry} \implies \frac{\sqrt{n}(\mu_2 - \mu_1)}{\sigma\sqrt{2}} = z_{1-\alpha} + z_{1-\beta}$$

$$\sqrt{n} = (z_{1-\alpha} + z_{1-\beta}) \cdot \left(\frac{\sigma\sqrt{2}}{\mu_2 - \mu_1}\right)$$

$$n = (z_{1-\alpha} + z_{1-\beta})^2 \cdot \left(\frac{\sigma\sqrt{2}}{\mu_2 - \mu_1}\right)^2$$

$$\boxed{n = 2(z_{1-\alpha} + z_{1-\beta})^2 \cdot \frac{\sigma^2}{(\mu_2 - \mu_1)^2}}$$

Final Sample Size

Because there are n observations per group (X_1, ..., X_n and Y_1, ..., Y_n), the total sample size is

$$4(z_{1-\alpha} + z_{1-\beta})^2 \cdot \frac{\sigma^2}{(\mu_2 - \mu_1)^2}$$